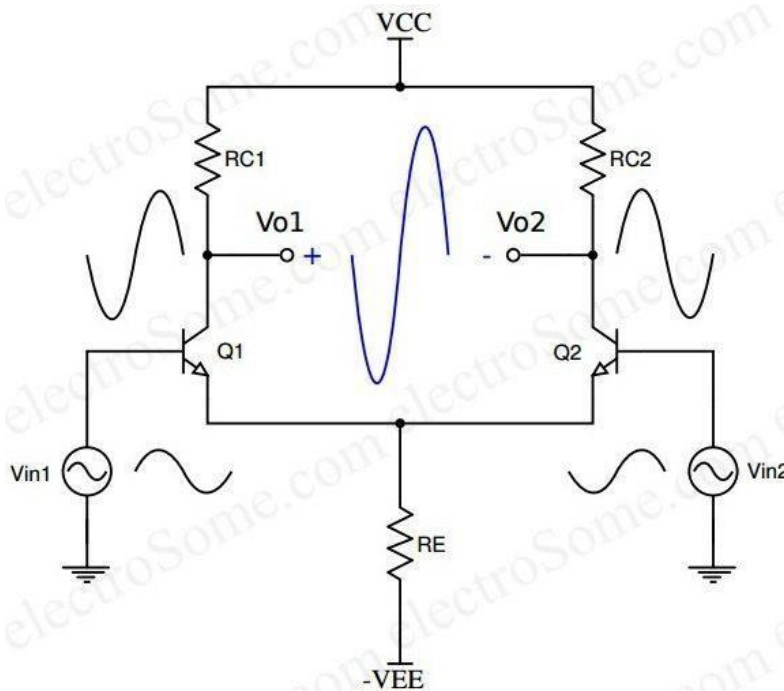


Differential amplifier using transistor CMRR (Common mode and Differential modes)

Aim :

This experiment aims to design a differential amplifier using transistor and to find out the Common Mode Rejection Ratio (CMRR).

THEORY



Above circuit consists of two identical transistors Q1 and Q2 with its emitters coupled together. Collectors are connected to main supply V_{CC} through collector resistor R_c . Magnitude of power supplies V_{CC} and $-V_{EE}$ will be same.

- $V_o = A_d(V_{in1} - V_{in2})$

Where A_d = differential gain

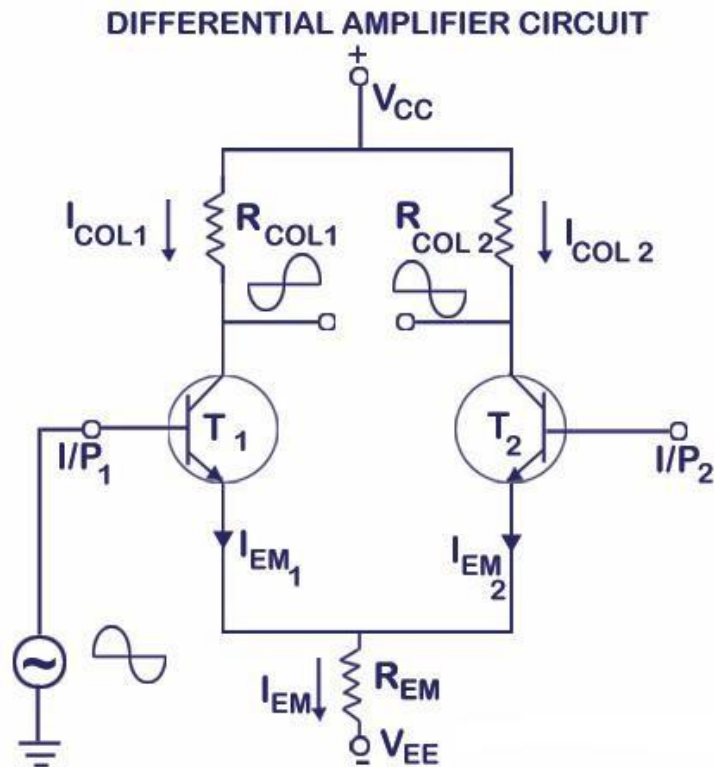
V_{in1} , V_{in2} = input voltages

When $V_{in1} = V_{in2}$, obviously the output will be zero. ie, differential amplifier suppresses common mode signals.

For effective operation, components on either sides should be match properly. Input signals are applied at base of each transistor and output is taken from both collector terminals. There won't be any unnecessary dc content in balanced output as the dc contents in both outputs gets cancelled each other.

Working of a Differential Amplifier

When a differential amplifier is driven at one of the inputs, the output appears at both the collector outputs. This is explained with a diagram below.



When input signal I/P_1 is applied to the transistor T_1 , there will be a high voltage drop across the collector resistance R_{COL1} , and thus the collector of T_1 will be less positive. When I/P_1 is negative T_1 is turned OFF, and the voltage drop across R_{COL1} becomes very low and thus the collector of T_1 will be more positive. Thus we can conclude that an inverted output appears at T_1 's collector for applying signal at I/P_1 .

When T_1 is turned ON by the positive value of I/P_1 , the current through the emitter resistance R_{EM} increases as the emitter current is almost equal to the collector current ($I_{E} \approx I_{C}$). Thus the voltage drop across R_{EM} increases and makes the emitter of both transistors go in a positive direction. Making T_2 's emitter positive is the same as making the base of T_2 negative. In such a condition the transistor T_2 will conduct less current which in turn will cause less voltage drop in R_{COL2} and thus the collector of T_2 will go in a positive direction for positive input signal. Thus we can conclude that the non-inverting output appears at the collector of transistor T_2 for input at base of T_1 .

The amplification can be driven differentially by taking output between the collector of T_1 and T_2 .

As shown in the figure above, if the transistors T_1 and T_2 are assumed to be identical in all characteristics, and if the voltages are equal ($V_{BASE1} = V_{BASE2}$), then the emitter current can also be said to be equal

$$I_{EM1} = I_{EM2}$$

$$\text{Total Emitter Current, } I_E = I_{EM1} + I_{EM2}$$

$$V_{EM} = V_{BASE} - V_{BASE-EM}$$

$$I_{EM} = (V_{BASE} - V_{BASE-EM})/R_{EM}$$

The emitter current I_{EM} remains virtually constant regardless of the h_{fe} value of the transistors.

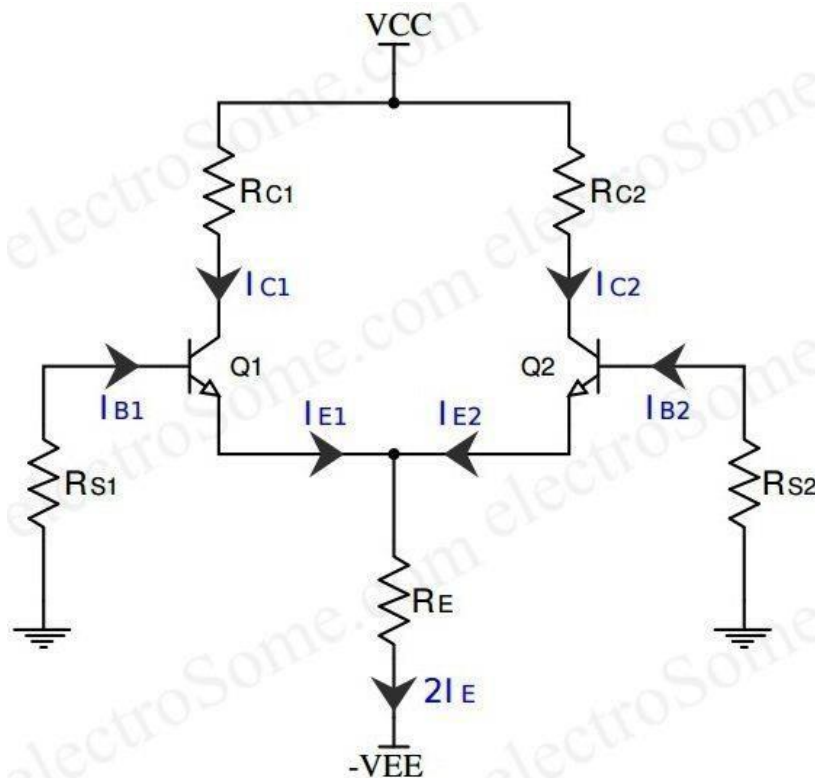
Since $I_{COL1} \approx I_{EM1}$, and $I_{COL2} \approx I_{EM2}$, $I_{COL1} \approx I_{COL2}$

Also, $V_{COL1} = V_{COL2} = V_{CC} - I_{COL} R_{COL}$, assuming collector resistance $R_{COL1} = R_{COL2} = R_{COL}$.

Differential amplifier is a closed loop amplifier circuit which amplifies the difference between two signals. Such a circuit is very useful in instrumentation systems. Differential amplifiers have high common mode rejection ratio (CMRR) and high input impedance.

DC Analysis

DC analysis provides the operating point values I_{CQ} and V_{CEQ} for the transistors used in the circuit. The DC equivalent circuit obtained by reducing all AC signals to zero as shown in figure below.



Assume :

- Source Resistance, $R_{S1} = R_{S2} = R_S$
- $Q1 = Q2$
- $R_{E1} = R_{E2}$ hence $R_E = R_{E1} || R_{E2}$
- $R_{C1} = R_{C2} = R_C$
- $|V_{CC}| = |V_{EE}|$

Applying KVL to base – emitter loop of Q1,

- $-I_B R_S - V_{BE} - 2I_E R_E + V_{EE} = 0$ —————(1)

Since $I_C = \beta I_B$ and $I_C \approx I_E$

$$\bullet \quad I_B \approx I_E / \beta \quad \text{-----}(2)$$

Substituting (2) in (1) =>

$$\begin{aligned} \bullet \quad & -I_E R_S / \beta - V_{BE} - 2I_E R_E + V_{EE} = 0 \\ \bullet \quad & -I_E (R_S / \beta - 2R_E) + V_{EE} - V_{BE} = 0 \end{aligned}$$

Where $V_{BE} = 0.7V$ for silicon and $0.2V$ for germanium.

Also $R_S / \beta \ll 2R_E$

Therefore,

$$\bullet \quad I_E = (V_{EE} - V_{BE}) / 2R_E \quad \text{-----}(3)$$

Eqn.(3) =>

- R_E determines the emitter current of Q1 and Q2
- I_E is independent of R_C

Since the voltage drop across R_S is too small, we can neglect it. So,

$$\begin{aligned} \bullet \quad & V_E = -V_{BE} \\ \bullet \quad & V_C = V_{CC} - I_C R_C \end{aligned}$$

$$\begin{aligned} V_{CE} &= V_C - V_E \\ &= (V_{CC} - I_C R_C) - (-V_{BE}) \\ &= V_{CC} + V_{BE} - I_C R_C \quad \text{-----}(4) \end{aligned}$$

We can determine operating point values using equations (3) and (4). Since two transistors are identical, same equations can be used for both.

$$\begin{aligned} \bullet \quad & I_E = I_C = (V_{EE} - V_{BE}) / 2R_E \\ \bullet \quad & V_{CE} = V_{CC} + V_{BE} - I_C R_C \end{aligned}$$

Differential mode gain:

Differential mode gain is given by the expression, $A_d = \frac{A_1 - A_2}{2}$, where A_1

and A_2 are the individual gains of the CE amplifiers. AC analysis of the circuit gives

$$A_d = -\frac{R_C}{2r_e} \text{ for a single ended output. If the input signal is applied at the base of the}$$

transistor T1 and the base of T2 grounded collector voltage of T1 drops and the collector voltage of T2 rises by the same amount.

Common mode gain:

Common mode gain represents the gain of the differential amplifier when the same input is fed to both base terminals. AC analysis of the circuit gives

$$A_c = -\frac{R_C}{2R_E} \text{ for a single ended output.}$$

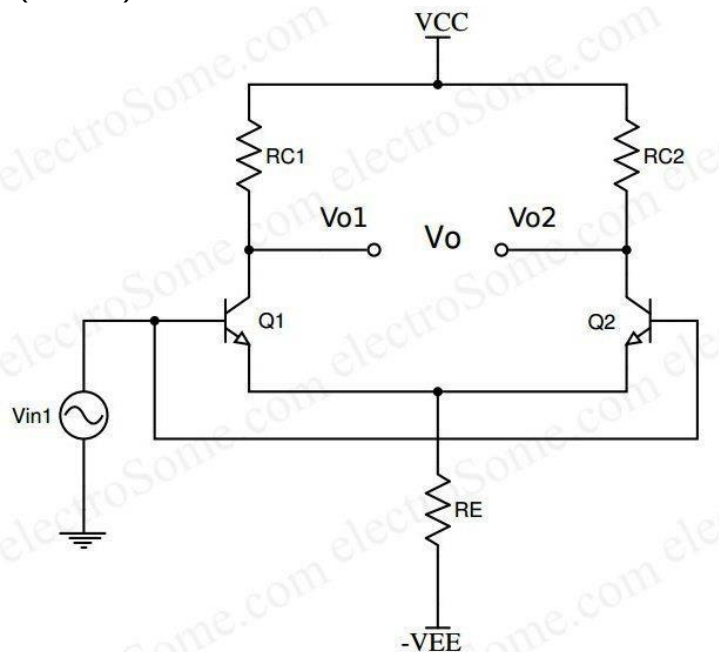
The input signals are applied in phase to bases

of transistors T1 and T2 so that both collector currents are equal in magnitude and direction. Hence the output will be balanced and approximately zero.

Common Mode Rejection Ratio (CMRR):

Dual input balanced output differential amplifier should suppress the common signals present at its inputs. A differential amplifier is said to be in common mode when same signal is applied to both inputs and the expected output will be zero, ie ideally common mode gain is zero.

Effectiveness of rejection depends on the matching of two common – emitter stages used. The ability of a differential amplifier to reject common mode signal is called Common Mode Rejection Ratio (CMRR).



$$\begin{aligned} \text{CMRR} &= \text{differential mode gain / common mode gain} \\ &= (A_d / A_c) \\ &= 20 \log(A_d / A_c) \quad (\text{usually CMRR is expressed in dB}) \end{aligned}$$

As mentioned earlier, ideally output will be zero in common mode which implies infinite CMRR.

Design:

For example, $V_{CC} = -V$; $V_{EE} = -V_{CC}$ V

$V_{CE} = \text{---- \% of } V_{CC}$ ie, $V_{CE} = -V$

$I_C = I_E = 1 \text{ mA}$

Design of R_C :

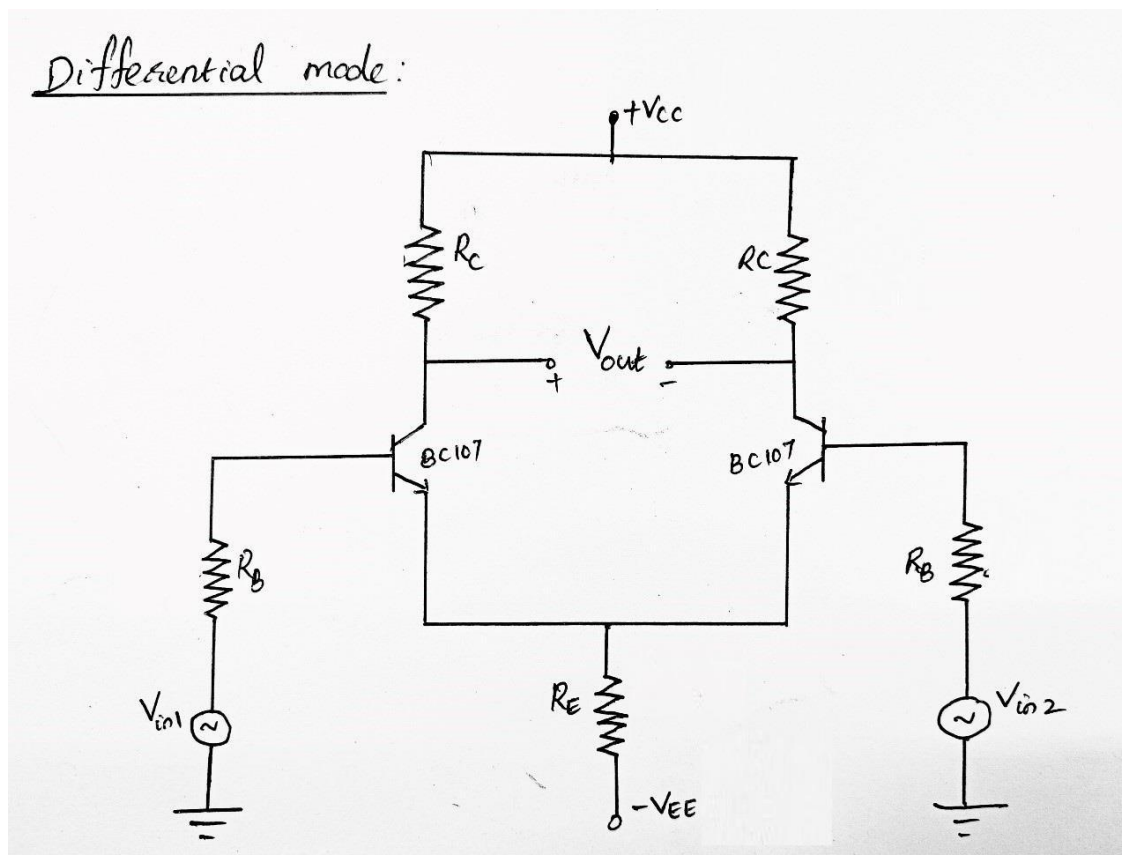
Assume ---- \% of V_{CC} drops across R_C , therefore $V_{RC} = \text{-----} V$

$$R_C = \frac{V_{RC}}{I_C}$$

Design of R_E :

$$I_E = \frac{(V_{EE} - V_{BE})}{2R_E}$$

$$R_E = \frac{(V_{EE} - V_{BE})}{2I_E}$$

Circuit Diagram :

Differential mode Gain, $A_d = \frac{V_{out}}{V_{in}}$

Common Mode:

The diagram shows a differential amplifier circuit configured for common mode operation. Two identical NPN transistors are used. Their emitters are connected to a common emitter resistor R_E , which is connected to the negative supply $-V_{EE}$. The bases of both transistors are connected to a common input signal V_{in} through a resistor. The collectors are connected to the positive supply $+V_{EE}$ through collector resistors R_C . The output V_{out} is taken from the collector of the right transistor.

$$CMRR = \left| \frac{A_d}{A_c} \right|$$

Result:

A differential amplifier was designed using transistor and the Common Mode Rejection Ratio (CMRR) was found to be __ **dB**.